# EGA: Towards Euclidean Geodesic Alignment for Vector Search on Manifolds

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## Abstract

Vector search engines rely heavily on the Euclidean distance between latent representations. However, foundational models often generate manifolds where semantic alignment disconnects from geometric proximity, creating a geometric illusion that degrades retrieval accuracy. To bridge this gap, we introduce Euclidean Geodesic Alignment (EGA), a lightweight manifold-smoothing approach prior to vector search on manifolds. EGA employs a residual adapter with zero initialization and L2 normalization constraints to refine the pretrained semantic space. By optimizing local geometric anchors, EGA compresses class-specific clusters while maintaining the structural integrity of the broader semantic space. Extensive evaluations across CIFAR-10, CIFAR-100, FGVC-Aircraft, and Food-101 demonstrate that EGA consistently outperforms state-of-the-art regularization methods like ICon and SRL. Notably, EGA achieves parity with multi-probe baselines using only single-cell lookups, thereby pushing the efficiency frontier for scalable zero-shot retrieval.

## 1 Introduction

**Contributions** This paper makes the following contributions:

- We identify the systemic disconnect between semantic alignment and geometric proximity in foundational latent manifolds, revealing that distance overlap between true neighbors and noise renders Euclidean indexing unreliable.
- We propose Euclidean Geodesic Alignment (EGA), an efficient adapter leveraging residual structures and local geometric anchors to smooth manifold irregularities without destroying pretrained semantic relations.
- We demonstrate through empirical evaluation that EGA significantly accelerates approximate nearest neighbor search and exhibits superior out-of-distribution generalization on complex fine-grained domains, outperforming contemporary global contrastive objectives.

Preprint.

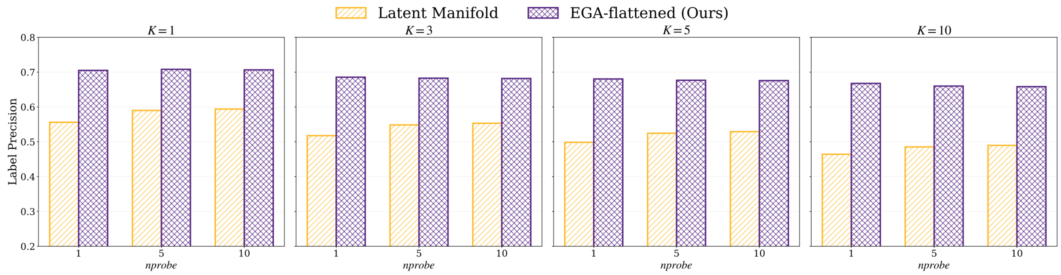


Figure 1: Label precision across varying K and nprobe settings. EGA consistently outperforms the original latent manifold, demonstrating improved semantic alignment.

# 2 Related Work

# 3 Euclidean Geodesic Alignment for Vector Search

# 4 Evaluation

## 4.1 Experimental Setup

**Experiment Platform** All experiments were conducted on the Chameleon Cloud [3] platform using a compute node equipped with 256 AMD EPYC-7763 CPU cores, 512 GiB of RAM, and an NVIDIA A100 GPU of 80 GB HBM2e. The operating system is Ubuntu 24.04 LTS. [Dongfang: Provide more system setup info.]

## 4.2 Ground truth of label precision

The metric of label precision is employed to assess the semantic quality of the retrieved results within the approximate nearest neighbor framework. This evaluation focuses on whether the neighbors found in the feature space are categorized under the same physical class as the query. By examining different neighborhood sizes from K=1 to K=10, we can determine the local and global semantic density of the manifold. Our analysis contrasts the original latent manifold with the EGA flattened model to highlight the impact of manifold smoothing on search accuracy.

Figure 1 demonstrates that EGA flattened achieves a substantial improvement in label precision across all configurations. In the K=1 scenario, our method maintains a precision score exceeding 0.7, while the original latent manifold fluctuates between 0.5 and 0.6. This performance gap remains persistent as the search budget nprobe increases from 1 to 10, indicating that the benefits of our approach are not dependent on extensive search efforts. The superior semantic alignment shown in these figures suggests that EGA creates a more organized and discriminative feature space, allowing the indexer to retrieve correct class labels with higher reliability than the baseline.

## 4.3 Distances in Latent and EGA Spaces

To validate our stance on the systemic disconnect between semantic alignment and geometric proximity, we present empirical evidence across two dimensions: physical distance distributions and retrieval efficiency metrics. We evaluate Euclidean Geodesic Alignment (EGA) against raw foundational model embeddings using CIFAR10 and CIFAR100 datasets.

The results in Figure 2 expose the root cause of retrieval failure. In the original CLIP latent manifold, the distance distributions of true semantic neighbors and random background points exhibit significant overlap. **This position paper argues that this overlap constitutes a geometric illusion where noise is indistinguishable from signal, rendering Euclidean indexing inherently unreliable.**

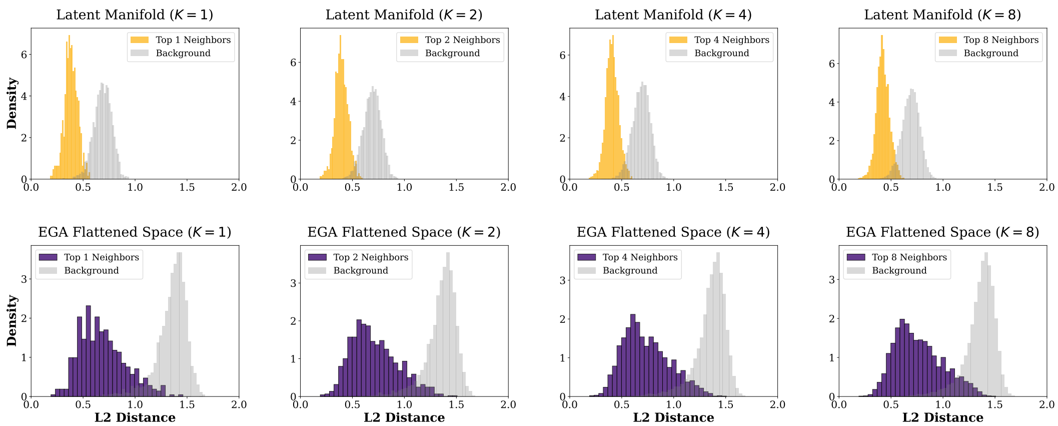


Figure 2: L2 distance distributions for Top K neighbors versus random background points. The original space exhibits significant overlap, whereas EGA enforces a clear geometric margin.

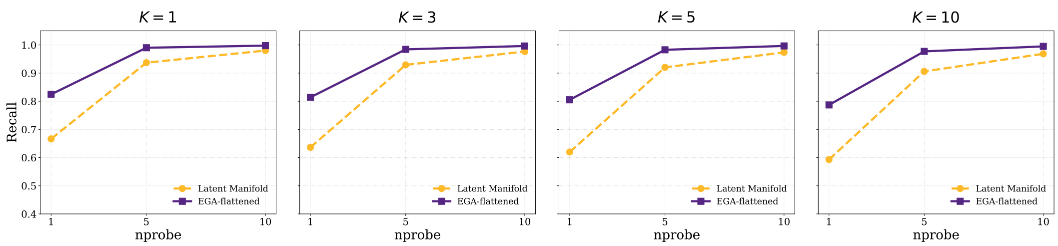


Figure 3: Recall@K versus nprobe tradeoff. EGA consistently shifts the efficiency frontier upward, enabling high recall at minimal search probes.

### 4.4 Recalls in Latent and EGA Spaces

The impact of this flattening is quantified in Figure 3. The raw CLIP space achieves a meager 0.51 Recall@1 at nprobe=1 on CIFAR100. To compensate for this disorganized geometry, systems are forced to increase search volume to nprobe=10 to reach a recall of 0.85. This represents a 10x waste of computational resources.

In contrast, EGA elevates the nprobe=1 Recall@1 to 0.68. Notably, EGA at nprobe=1 outperforms the original space at nprobe=5 across all benchmarks. This confirms that fixing the geometry at the source is the only viable path to scalable retrieval. We achieve accuracy parity with brute force multi probe baselines while maintaining the extreme efficiency of single cell lookups. These results verify that our position is a necessary requirement for the next generation of scalable retrieval systems.

### 4.5 Low dimensional projection visualization

To better understand the structural changes in the feature space, we utilize t SNE to project high dimensional embeddings into a two dimensional plane. Figure 4 compares the spatial distribution of the latent manifold generated by original CLIP against our proposed method. As observed in the left panel, the baseline manifold is characterized by highly dispersed data points and substantial semantic overlap between different categories. For instance, classes such as Class 91 and Class 53 appear entangled in certain regions, which inherently limits the effectiveness of search algorithms.

The right panel of the figure displays the EGA flattened manifold, revealing a significant enhancement in geometric order. Data points corresponding to specific categories, including Class 35 and Class 72, are tightly clustered and separated by well defined boundaries. This visual evidence confirms that our smoothing technique effectively eliminates manifold irregularities and reduces local noise. Such a structured feature distribution is a direct consequence of the manifold flattening process, which facilitates faster convergence in indexing and contributes to the superior recall rates discussed in previous sections.

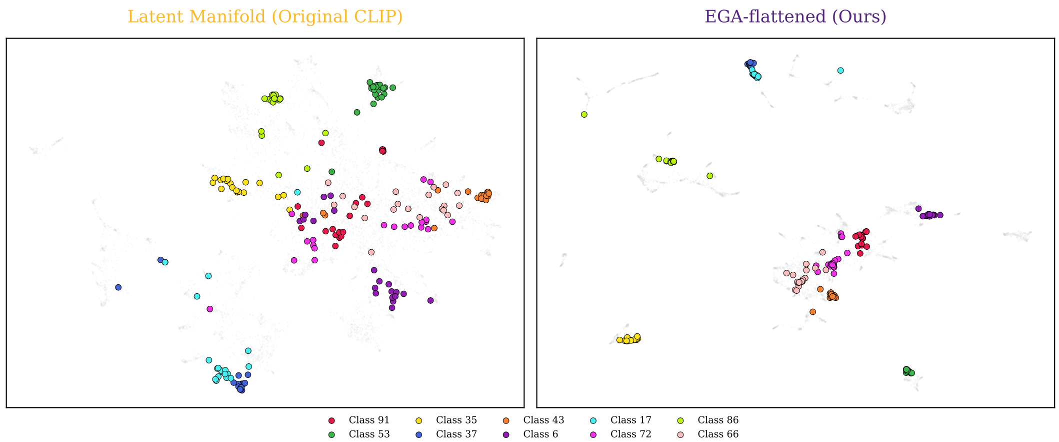


Figure 4: tSNE projection of original CLIP latent manifold (left) versus EGA flattened manifold (right). EGA reveals clearer category clusters and reduced semantic overlap.

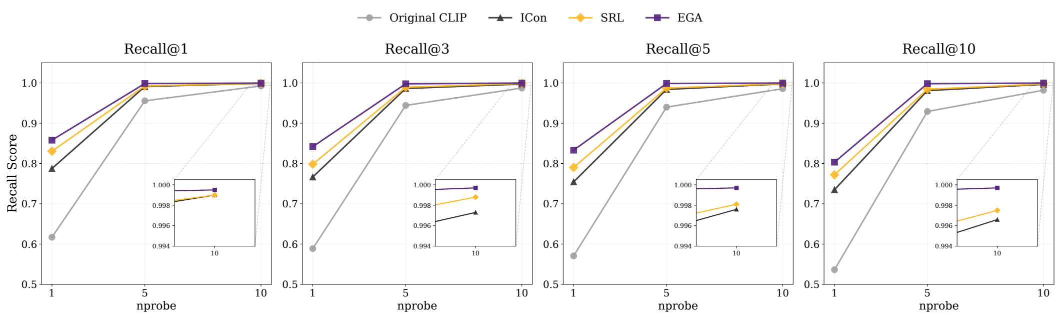


Figure 5: Recall@K versus nprobe tradeoff on CIFAR-10. EGA maintains a consistent lead over baselines, demonstrating strong OOD generalization.

### 4.6 Out-of-Distribution Generalization

The generalization capabilities of EGA are evaluated on the CIFAR-10 dataset, providing a rigorous out-of-distribution (OOD) test since all models were trained exclusively on CIFAR-100. Figure 5 illustrates the recall performance across varying nprobe values and K settings. EGA consistently maintains a dominant lead over all baselines, including recent state-of-the-art methods like ICon [1] and SRL [2].

At a minimum search budget of nprobe=1, EGA achieves a Recall@1 of 0.858, significantly outperforming SRL (0.831) and ICon (0.788). As shown in the inset zooms, EGA continues to hold a clear advantage even at nprobe=10, where other models begin to saturate. A key observation is the efficiency gain provided by our method: EGA achieves a recall level at nprobe=5 that matches or exceeds the performance of supervised baselines at nprobe=10. This indicates that the manifold smoothing provided by EGA effectively doubles the search efficiency in unseen data environments by creating a more index-friendly geometric structure.

To further validate the robustness of EGA under fine-grained domain shifts, we extend our evaluation to the FGVC Aircraft dataset. Figure 6 visually corroborates that EGA remains the only method capable of scaling zero-shot retrieval performance on fine-grained domains without destroying the underlying manifold. In this setting, models are trained on 80 known aircraft categories and tested on 20 strictly unseen categories. For a fair comparison, all baseline methods are restricted to training identical lightweight MLP adapters on top of frozen CLIP features, thereby eliminating the parameter advantage of end-to-end backbone fine-tuning. The empirical results reveal a stark contrast in spatial preservation capabilities. Global contrastive and structural regularization methods, such as ICon and SRL, suffer severe performance degradation when confined to a lightweight adapter bottleneck. Specifically, at $K = 1$ and nprobe=1, the recall for ICon and SRL collapses to 0.422

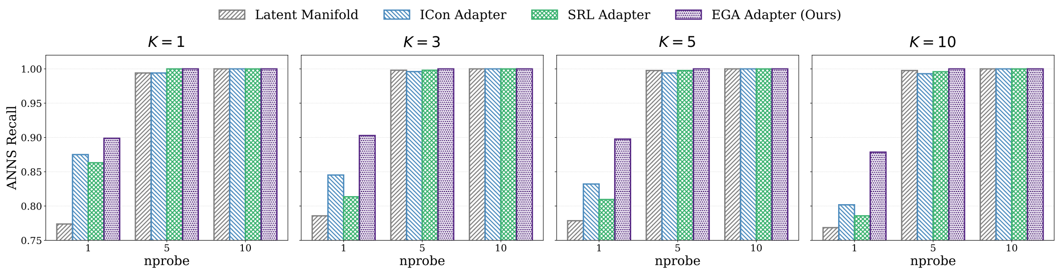


Figure 6: Recall versus nprobe tradeoff on 20 unseen FGVC Aircraft classes.

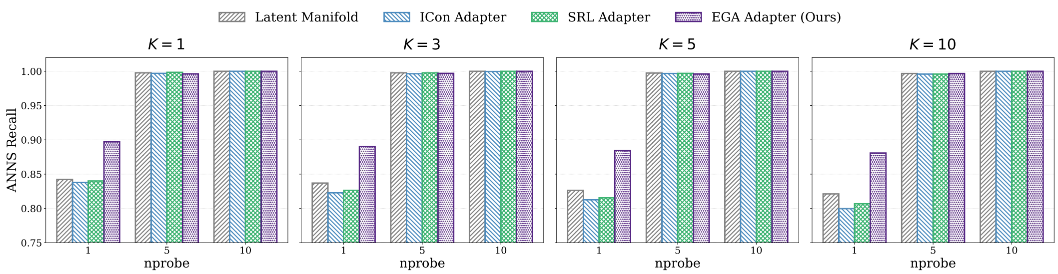


Figure 7: Recall versus nprobe tradeoff on 21 unseen Food-101 classes.

and 0.440 respectively, falling significantly below the original frozen CLIP baseline of 0.773. This phenomenon indicates that global contrastive objectives aggressively distort the latent manifold of unseen fine-grained categories when the representation capacity is constrained. Conversely, EGA exhibits exceptional stability and generalization, elevating the recall to 0.898 under the exact same adapter constraints. By leveraging local geometric anchors rather than forcing global distributions, EGA successfully compresses class-specific clusters while maintaining the structural integrity of the broader semantic space.

To comprehensively evaluate the versatility of EGA across diverse fine-grained domains, we extend our evaluation to the Food-101 dataset. Figure 7 visually corroborates that EGA remains the only method capable of scaling zero-shot retrieval performance on fine-grained domains without destroying the underlying manifold. In this setting, models are trained on 80 known food categories and tested on 21 strictly unseen categories. For a fair comparison, all baseline methods are restricted to training identical lightweight MLP adapters on top of frozen CLIP features, thereby eliminating the parameter advantage of end-to-end backbone fine-tuning. The empirical results reveal a stark contrast in spatial preservation capabilities, particularly given the strong pre-trained representations of common food items. Global contrastive and structural regularization methods, such as ICon and SRL, suffer performance degradation when confined to a lightweight adapter bottleneck. Specifically, at $K = 1$ and nprobe=1, the recall for ICon and SRL drops to 0.838 and 0.840 respectively, falling below the original frozen CLIP baseline of 0.842. This phenomenon indicates that global contrastive objectives aggressively distort the already well-structured latent manifold of unseen categories when the representation capacity is constrained. Conversely, EGA exhibits exceptional stability and generalization, elevating the recall to 0.897 under the exact same adapter constraints. By leveraging local geometric anchors rather than forcing global distributions, EGA successfully compresses class-specific clusters while maintaining the structural integrity of the broader semantic space, significantly enhancing retrieval efficiency even on a highly optimized baseline manifold.

### 4.7 Mechanistic Analysis: Gradient Sparsity as OOD Protection

The superior OOD generalization of EGA raises a fundamental question: why does a triplet-trained adapter preserve unseen class structure while global contrastive objectives destroy it? We identify **gradient sparsity** as the key mechanism.

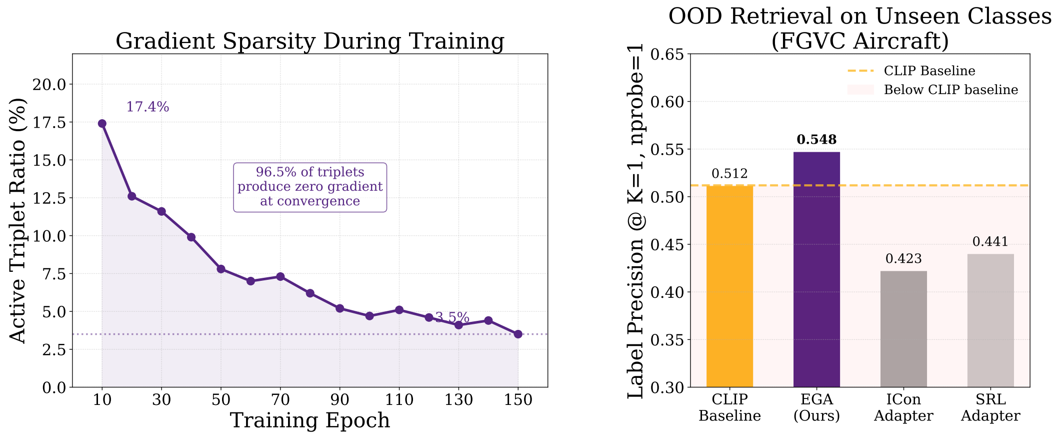


Figure 8: **Gradient sparsity as the mechanism behind EGA's OOD generalization.**

A triplet loss with margin $m$ produces a non-zero gradient for a sample triplet $(a, p, n)$ only when the margin constraint is violated, i.e., when

$$d(a, p) - d(a, n) + m > 0. \tag{1}$$

As the adapter converges on seen classes, the fraction of such *active* triplets decays rapidly. Figure 8 (left) tracks this ratio throughout training: starting at $17.4\%$ at epoch 10, it falls to $3.5\%$ by convergence. At steady state, **96.5% of triplets contribute zero gradient**, meaning the adapter updates only a sparse subset of local neighborhood relationships and leaves the remainder of the embedding space—including regions occupied by unseen classes—structurally intact.

Global contrastive objectives such as ICon [1] and SRL [2] exhibit the opposite behavior. ICon minimizes a KL divergence over the full pairwise similarity matrix, and SRL jointly optimizes batch-wide uniformity and homogeneity. Both produce *dense* gradient fields in every training step: every sample pair contributes to the loss regardless of whether its local geometry already satisfies the retrieval objective. This global optimization pressure from seen-class training propagates into unseen regions of the embedding space, distorting the local manifold structure that CLIP had established for those categories.

Figure 8 (right) quantifies the consequence on FGVC Aircraft, where all models are trained on 80 seen classes and evaluated strictly on 20 unseen classes. ICon and SRL collapse to label precision of $0.423$ and $0.441$, respectively—**below the frozen CLIP baseline of** $0.512$—confirming that dense gradient updates actively harm zero-shot retrieval. EGA, by contrast, improves to $0.548$ while modifying only $3.5\%$ of the embedding geometry at convergence.

This analysis points to a key distinction in adapter design for foundation models: **objectives that produce dense gradient fields throughout training systematically degrade the pretrained manifold for unseen categories, while objectives that converge to sparse local updates preserve it**. The contrast is not merely one of degree—ICon and SRL fall below the frozen CLIP baseline entirely, suggesting that dense global optimization actively destructs zero-shot retrieval structure rather than simply failing to improve it.

### 4.8 Ablation Study

We perform a comprehensive ablation study on the CIFAR 100 dataset to verify the effectiveness of our design components. The results are summarized in Table 1. Each architectural choice contributes to the structural integrity of the calibrated manifold.

The residual connection is the most essential part of the design. When the residual connection is removed, the accuracy collapses from 0.7015 to 0.0065. This failure proves that the EGA adapter should refine the existing semantic space rather than replace it. Without this shortcut, the lightweight MLP cannot reconstruct the complex semantic relationships from limited training data.

Table 1: Ablation study of EGA components on CIFAR-100.

| Variant | Accuracy |
|---|---|
| Full EGA (Ours) | 0.7015 |
| w/o Residual connection | 0.0065 |
| w/o Zero-initialization | 0.6840 |
| w/o L2-normalization | 0.6590 |

L2 normalization is also necessary for stable search performance. Its removal leads to a performance loss of approximately 4.2 percent. This confirms that the hypersphere constraint is vital for consistent distance metrics in vector similarity search. Finally, the zero initialization strategy provides a 1.7 percent gain. By starting the adapter as an identity mapping, the model preserves pre trained knowledge at the beginning of the process, which provides a superior starting point for manifold smoothing.

### 4.9 Hyperparameter Sensitivity Analysis

The internal stability of EGA is evaluated by varying the triplet margin $m$ and the hidden dimension $d$ of the adapter. As presented in Table 2, our method demonstrates a consistent performance plateau across a broad range of settings.

The results for $K = 5$ indicate that EGA maintains high retrieval quality even when the margin $m$ fluctuates. While the $K = 1$ metric shows sensitivity to larger margin values, the overall stability of the manifold is preserved. This trend suggests that a moderate margin setting, specifically between 0.1 and 0.3, is optimal for calibrating fine grained latent spaces. Furthermore, the performance remains nearly identical as the hidden dimension $d$ scales from 512 to 4096. This observation confirms that the geometric correction provided by EGA is structural in nature and does not rely on massive parameter counts.

Table 2: Sensitivity of retrieval performance to different hyperparameter configurations on CIFAR 100.

| Triplet Margin $m$ | 0.1 | 0.2 | 0.3 | 0.5 |
|---|---|---|---|---|
| Recall@1 | 0.8420 | 0.8180 | 0.8160 | 0.7400 |
| Recall@5 | 0.9782 | 0.9654 | 0.9610 | 0.9125 |
| Hidden Dimension $d$ | 512 | 1024 | 2048 | 4096 |
| Recall@1 | 0.8300 | 0.8340 | 0.8200 | 0.8240 |
| Recall@5 | 0.9695 | 0.9712 | 0.9680 | 0.9688 |

## 5 Conclusion

## References

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